

Cryptography - 12_12

Identity-Based Encryption

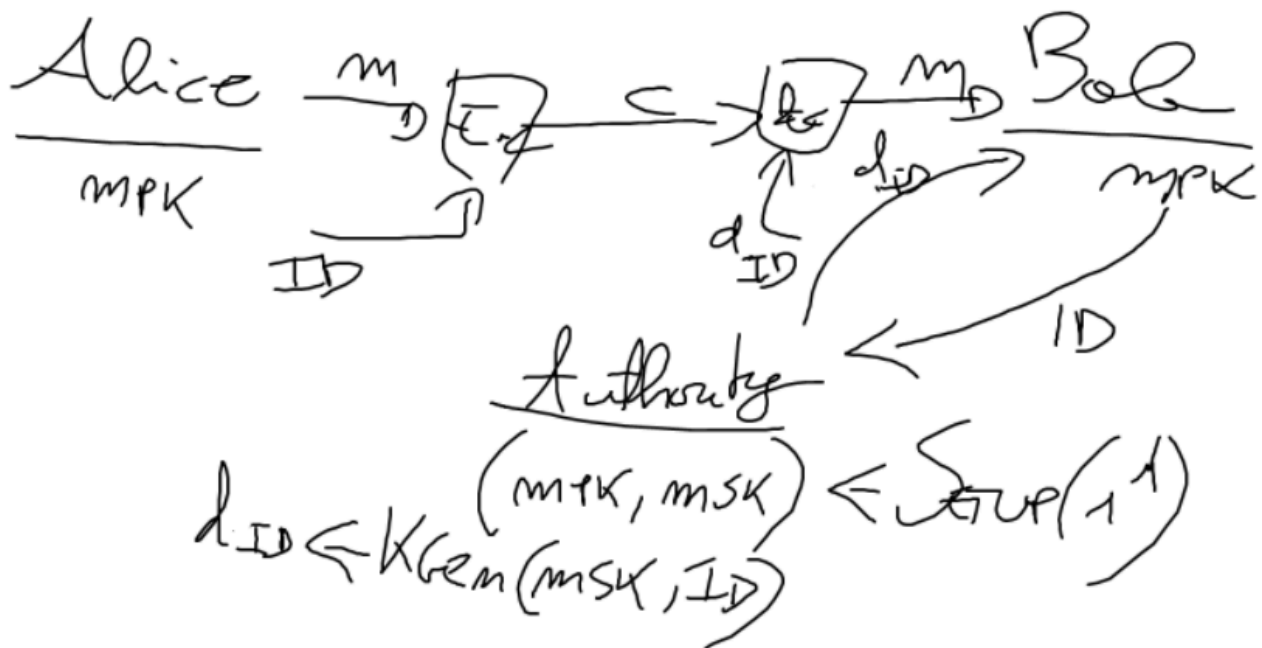
IBE: Public-Key encryption where the pk can be arbitrary string.

Why is it useful? If we can associate unique identities to people (e.g. social security number), we can encrypt only knowing an $ID \in \{0, 1\}^*$.

Since we need to generate secret keys, we need an authority anyways.

Otherwise, everybody could generate secret keys for arbitrary identities.

Syntax: $\Pi = (Setup, KGen, Enc, Dec)$.



Basically the Authority is able to generate two keys, (mpk, msk) (master public and secret key), and everyone knows mpk . Now, when Bob wants to get his secret key, he sends his id to the Authority, who computes d_{id} by using $KGen(msk, id)$ and sends it to Bob.

Now, Alice will encrypt the message m using the id , producing a ciphertext c , which can be decrypted using d_{id} .

Key escrow: It's a new issue. The authority can generate all secret keys. So, it can decrypt everything.

Advantage: No certificates. Zero overhead.

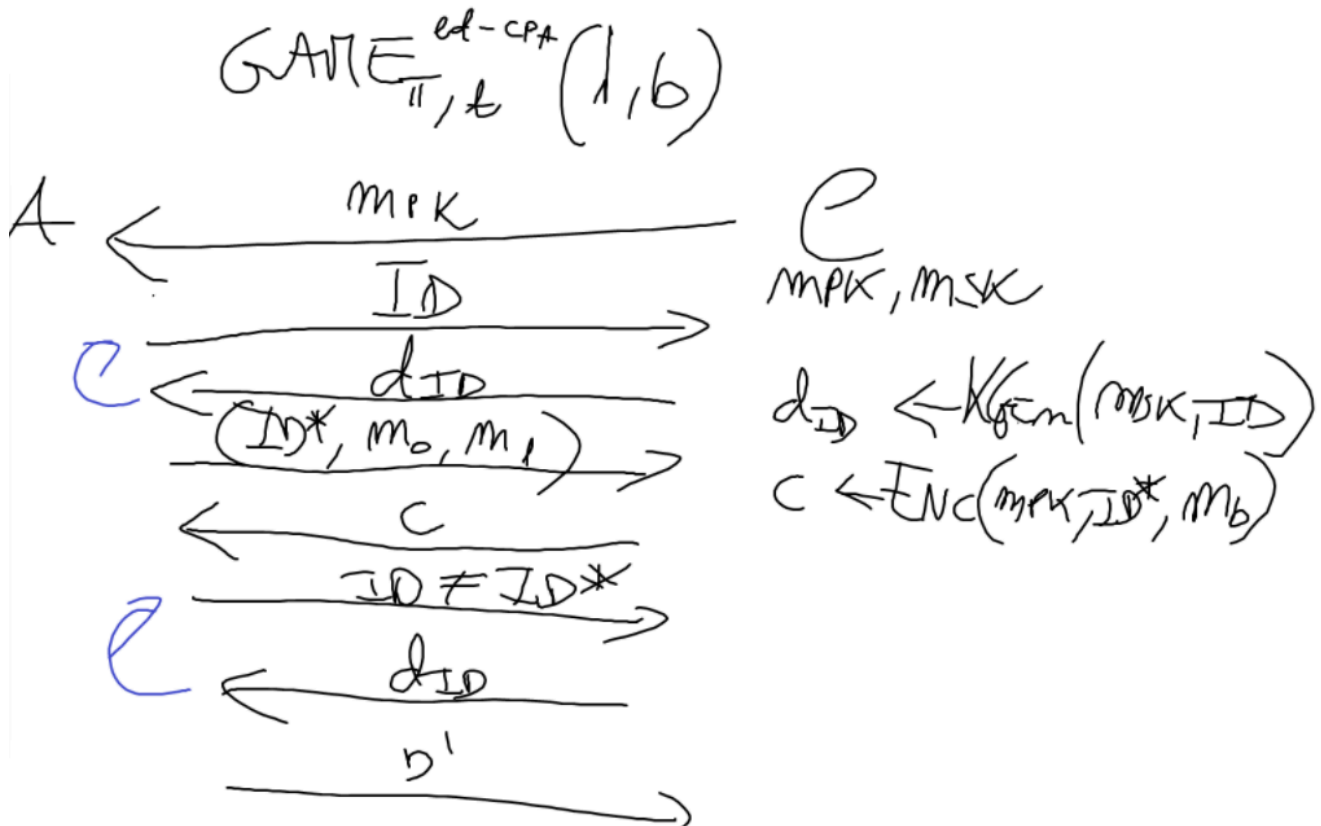
Why is it interesting?

- A generalization of PKE. With further generalizations one gets new forms of

encryption that are more expressive than PKE: ABE, Functional encryption.

- We will show that IBE implies both: Digital signatures, CCA-secure public key encryption.

Security: IND-ID-CPA security.



Note that $ID^* \neq \{ID\}$.

We're trying to break Bob in the scheme above.

Next goal: A construction of IBE.

This construction will satisfy a weaker definition. This variant is called **Selective IND-ID-CPA**. This means $\mathcal{A}$ picks ID^* before mpk is sampled.

Selective will be enough for us and the mentioned applications!

The construction: It will be based on so-called pairings. We will write:

$$(\mathbb{G}, \mathbb{G}_T, g, q, \hat{e}) \leftarrow \text{GroupGen}(1^\lambda)$$

Now:

1. $\mathbb{G}$ is cyclic with generator g and order q (prime).
2. $\hat{e} : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ (known as pairing) such that: $\forall g, h \in \mathbb{G}, \forall a, b \in \mathbb{Z}_q$

$$\hat{e}(g^a, h^b) = \hat{e}(g, h)^{ab}$$

($\mathbb{G}_T$ is also of prime order q).

Example: $\hat{e}$ exists in elliptic curve groups.

Consequence: DDH does not hold in bi-linear groups. Given g^a, g^b, g^c :

$$\hat{e}(g, g^c) = \hat{e}(g^a, g^b) \quad (= \hat{e}(g, g)^{ab})$$

Here is an assumption that is believed to hold:

✓ Definition (DBDH)

$$(g^a, g^b, g^c, \hat{e}(g, g)^{abc}) \approx_c (g^a, g^b, g^c, T)$$

for $a, b, c \leftarrow \mathbb{Z}_q, T \leftarrow \mathbb{G}_T$

The construction:

- $Setup(1^\lambda)$: $(\underbrace{\mathbb{G}, \mathbb{G}_T, g, q, \hat{e}}_{params}) \leftarrow GroupGen$.

Pick $h \leftarrow \mathbb{G}$, $g_2 \leftarrow \mathbb{G}$ and $g_1 = g^\alpha$ for $\alpha \leftarrow \mathbb{Z}_q$. Then:

$$mpk = (params, g, g_1, g_2, h)$$

$$msk = g_2$$

- $KGen(msk, ID)$: Consider $F : \mathbb{Z}_q \rightarrow \mathbb{G}$ such that $F(ID) = g_1^{ID} \cdot h$.

Given $ID \in \mathbb{Z}_q$, pick $r \leftarrow \mathbb{Z}_q$ and output $d_{ID} = (d_0, d_1)$, where

$$d_0 = \underbrace{g_2^\alpha}_{msk} \cdot F(ID)^r; d_1 = g^r.$$

(Using $H : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ we extend the $ID \in \{0, 1\}^*$)

- $Enc(mpk, ID, m)$: Given $m \in \mathbb{G}_T$; pick $\gamma \leftarrow \mathbb{Z}_q$ and output $c = (u, v, w)$ where $u = \hat{e}(g_1, g_2)^\gamma \cdot m$, $v = g^\gamma$, $w = F(ID)^\gamma$.
- $Dec(mpk, d_{ID}, c)$: Parse $c = (u, v, w)$ and $d_{ID} = (d_0, d_1)$. Then output $\frac{u \cdot \hat{e}(d_1, w)}{\hat{e}(v, d_0)}$.

The correctness:

$$\frac{u \cdot \hat{e}(d_1, w)}{\hat{e}(v, d_0)} = \frac{\hat{e}(g_1, g_2)^\gamma \cdot m \cdot \hat{e}(g^r, F(ID)^\gamma)}{\hat{e}(g^\gamma, g_2^\alpha \cdot F(ID)^r)} = \frac{\hat{e}(g_1, g_2)^\gamma \cdot m \cdot \cancel{\hat{e}(g, F(ID))^{r \cdot \gamma}}}{\underbrace{\hat{e}(g^\gamma, g_2^\alpha)}_{=\hat{e}(g^\alpha, g_2^\gamma)=\hat{e}(g_1, g_2)^\gamma} \cdot \cancel{\hat{e}(g^\gamma, F(ID)^r)}}$$

✓ Theorem

The above IBE is selective IND-ID-CPA under the DBDH assumption.